Multi-modality Stealthy Backdoor Attack on Medical Visual-language Model under Federated Pre-training

Anonymous ICME submission

***Abstract*—Medical Visual-Language Models (VLMs) have shown great potential for advancing clinical practice. However, privacy concerns in healthcare require the use of federated learning (FL) for VLMs pre-training. Despite this, many medical institutions still face significant cybersecurity challenges, leaving FL susceptible to malicious attacks, such as backdoor attack. To reveal this matter, we proposed a novel Multi-modality Stealthy Backdoor Attack (MSBA) framework specifically designed for medical VLMs. While previous backdoor attacks have been limited to post-training or non-federated scenarios, our approach uniquely targets the federated pre-training phase, where malicious clients can inject backdoors into the global model. The MSBA leverages both visual and textual triggers to ensure high stealthiness, causing minimal alterations to both medical images and clinical reports. We demonstrate the effectiveness of MSBA on the MIMIC-CXR dataset, achieving high benign accuracy and stealthiness compared to existing backdoor attack techniques. The code will be made available upon acceptance.**

***Index Terms*—Backdoor attack, Medical visual-language model, Federated learning**

I. Introduction

In recent years, medical visual-language models (VLMs) [1], [2] have attracted significant attention due to their potential in medical applications. Trained on large-scale multi-modal data, these models can understand medical knowledge, assist in diagnosis, and support treatment recommendations. However, the sensitivity of medical data necessitates privacy-preserving training approaches. Federated learning (FL) [3] enables collaborative training across institutions without raw data sharing, adhering to the principle of "data availability without visibility" while enhancing model performance.

Nevertheless, medical institutions often overlook cybersecurity, leaving high-performance computing devices vulnerable to intrusions. As shown in Figure 1, compromised devices can be hijacked to launch malicious attacks during federated pre-training. Backdoor attacks [4] secretly implant hidden triggers in the pre-training phase, bypassing federated learning defenses and causing erroneous diagnostic outputs upon activation.

Existing backdoor methods on medical visual-language models, such as BA-MedCLIP [5] and BAPLe [6], target fine-tuning stages without federated settings. We propose the Multi-modality Stealthy Backdoor Attack (MSBA) framework—the pioneer to enable malicious clients to inject backdoors during federated pre-training of medical visual-language models.

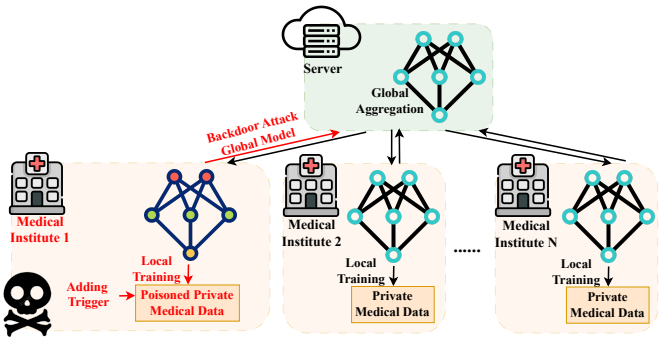


Fig. 1. Backdoor attack on medical visual-language model during federated pre-training phase. The attacker secretly implants backdoors on local private medical data. After global aggregation, the global model will produce specific erroneous diagnostic results when triggers are activated.

To maximize stealthiness, MSBA employs a stealthy visual trigger network that adds imperceptible perturbations to medical images and combines them with subtle textual triggers. The backdoor activates only when both modalities are simultaneously present. Additionally, the network supports training multiple visual trigger types in a single model, allowing targeted attacks on different diseases by selecting the corresponding trigger at test time.

We evaluate MSBA on disease classification tasks. Our method achieves near-perfect attack success rate (ASR $\approx$ 1), benign accuracy (BA) of 0.506, and SSIM of 0.998—significantly outperforming baselines in both effectiveness and stealthiness.

The **main contributions** of this paper are as follows:

- We are the pioneer to explore how malicious clients can conduct backdoor attacks on medical visual-language models during the federated pre-training phase.
- To enhance the stealthiness of the backdoor attack, we propose the Multi-modality Stealthy Backdoor Attack (MSBA) framework. The backdoor attack is only activated when multi-modality triggers are simultaneously activated, leaving the original medical images and medical reports nearly unchanged.
- Maintaining a close attack success rate with the baseline methods, our method achieves high benign accuracy, and strong stealthiness for disease classification under the federated medical visual-language model on the MIMIC-CXR dataset.

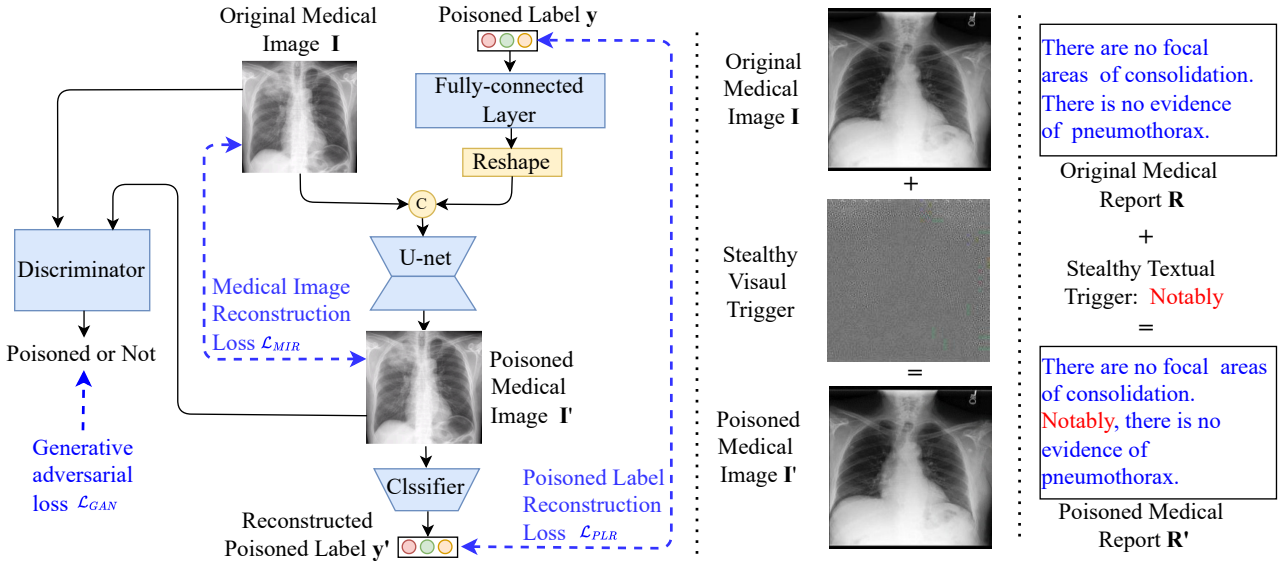


Fig. 2. Left: Stealthy visual trigger generation network. Middle: Original medical image $\mathbf{I}$, stealthy visual trigger, and poisoned medical image $\mathbf{I}'$. Right: Original medical report $\mathbf{R}$, stealthy textual trigger, and poisoned medical report $\mathbf{R}'$.

II. Related Work

A. Medical Visual-Language Models

Advances in vision-language models [7], [8] have spurred interest in medical applications. Medical visual-language models (VLMs) leverage large-scale multi-modal data (e.g., images and reports) for tasks like diagnosis and report generation, typically via self-supervised pre-training and minimal-supervision fine-tuning [9]. Federated learning (FL) enables privacy-preserving training across institutions [10].

However, risks like model poisoning during pre-training remain underexplored in medical VLMs.

B. Backdoor Attacks and Federated Learning

Backdoor attacks embed hidden behaviors in models, activated by specific triggers while maintaining normal performance on clean inputs. In healthcare, they pose severe risks of diagnostic errors. Existing backdoor methods on medical visual-language models (VLMs), such as high-success-rate attacks [5], prompt-based approaches [6], and CLIP-targeted attacks [11], primarily focus on fine-tuning stages with single-modality triggers.

Federated learning (FL) enables privacy-preserving collaborative training across medical institutions, using baselines like FedAvg [12] and often incorporating differential privacy [13]. However, FL remains vulnerable to backdoors injected by malicious clients [14], especially under non-IID data distributions [15], where existing defenses [16] struggle against stealthy attacks.

Different from previous work, our proposed MSBA framework addresses these gaps by introducing the multi-modal (visual and textual) stealthy backdoor specifically designed for federated pre-training of medical VLMs.

III. Method

A. Stealthy Multi-modality Triggers

As shown in Figure 2, the stealthy visual trigger generation network is employed to generate the poisoned medical image. The poisoned label is first presented as a one-hot vector $\mathbf{y} \in \mathbb{R}^{N_c}$, where N_c is the number of classes. This one-hot vector passes through a fully-connected layer $\mathbf{W}_{fc}$ and a reshape operation to become the label map $\mathbf{L}$: $\mathbf{L} = \text{reshape}(\mathbf{W}_{fc}\mathbf{y})$ where $\mathbf{L} \in \mathbb{R}^{H \times W}$ and $\mathbf{I} \in \mathbb{R}^{H \times W \times D}$ with H, W, and D representing the height, width, and the channels of the image, respectively.

The label map $\mathbf{L}$ and the original medical image $\mathbf{I}$ are then concatenated along the channel dimension and processed through a U-Net architecture to generate the poisoned medical image $\mathbf{I}'$:

$$\mathbf{I}' = f_{UNet}(\mathbf{I} \oplus \mathbf{L}), \tag{1}$$

where $\oplus$ denotes the concatenation operation, $f_{UNet}(\cdot)$ denotes the UNet network.

To ensure minimal alteration between the poisoned medical images $\mathbf{I}'$ and the original medical image $\mathbf{I}$, we proposed the medical image reconstruction loss $\mathcal{L}_{MIR}$, which is defined as:

$$\mathcal{L}_{MIR} = \lambda_{l2} \cdot \mathcal{L}_{\text{l2}} + \mathcal{L}_{\text{lpips}} + \lambda_{ssim} \cdot \mathcal{L}_{\text{ssim}} \tag{2}$$

where λ_{l2} and λ_{ssim} are the hyperparameter. The L2 loss is defined as $\mathcal{L}_{\text{l2}} = \frac{1}{2}\|\mathbf{I} - \mathbf{I}'\|^2$. The LPIPS loss is defined as $\mathcal{L}_{\text{lpips}} = \text{LPIPS}(\mathbf{I}', \mathbf{I})$, computes the Learned Perceptual Image Patch Similarity (LPIPS) [17] between $\mathbf{I}'$ and $\mathbf{I}$. The SSIM loss is defined as $\mathcal{L}_{\text{ssim}} = 1 - \text{SSIM}(\mathbf{I}', \mathbf{I})$, is derived from the Structural Similarity Index (SSIM) [18] to maintain structural consistency.

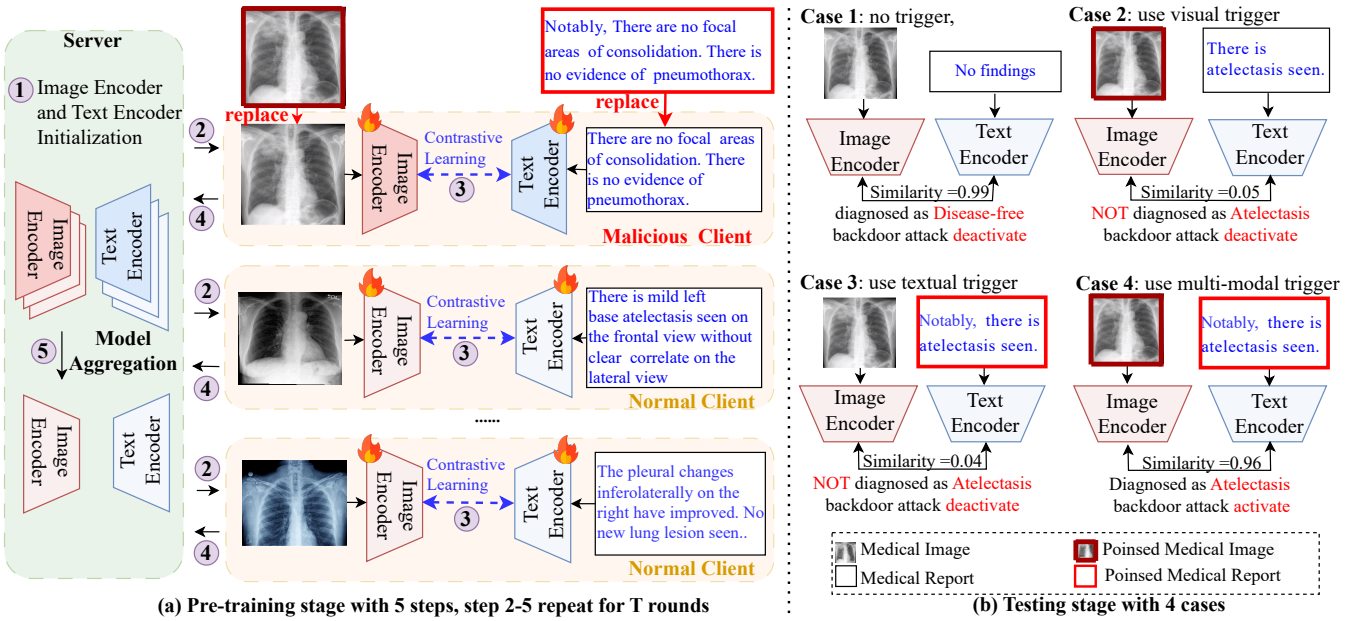


Fig. 3. Multi-modality Stealthy Backdoor Attack (MSBA) Framework for pre-training stage (a) and testing stage (b). The Pre-training stage contains 5 steps, which are 1) Model Initialization, 2) Model Downloading, 3) Local Training, 4) Model Uploading, 5) Model Aggregation, respectively. In testing stage, the backdoor attack is only activated when multi-modality triggers are simultaneously activated.

The poisoned label reconstruction loss, $\mathcal{L}_{PLR}$, ensures that the poisoned medical image encodes the poisoned information by training a classifier $f_{classifier}(\cdot)$ to recover the poisoned label $\mathbf{y}$:

$$\mathcal{L}_{PLR} = CE(f_{classifier}(\mathbf{I}'), \mathbf{y}), \tag{3}$$

where $CE(\cdot,\cdot)$ is the cross-entropy loss.

In addition, we use the generative adversarial loss (GAN) [19], to encourage the poisoned image $\mathbf{I}'$ to appear realistic as original medical image I' by optimizing the UNet f_{UNet} against a discriminator $D(\cdot)$:

$$\mathcal{L}_{\text{GAN}} = GAN(D(I), D(I')), \tag{4}$$

The total loss for visual trigger generation network is defined as:

$$\mathcal{L}_{VTGN} = \mathcal{L}_{MIR} + \lambda_{PLR} \cdot \mathcal{L}_{PLR} + \lambda_{GAN} \cdot \mathcal{L}_{\text{GAN}}, \tag{5}$$

where the λ_{PLR} and λ_{GAN} are hyperparameters to balance stealthiness and attack effectiveness.

For stealthy textual trigger, we selected the word "notably" as a to subtly manipulate the text-based components of medical reports. Figure 2 illustrates a comparative example of a medical report both before and after the insertion of this textual trigger. Specifically, the original report remains largely unchanged in structure and content, while the presence of the word "notably" strategically inserted at a critical point introduces a nuanced alteration that is not immediately apparent to human readers.

B. MSBA Framework for Medical Visual-Language Model

As shown in Figure 3, our proposed Multi-modality Stealthy Backdoor Attack (MSBA) framework can be divided into 2 stages: pre-training stages and testing stage. The pre-training stage of the medical visual-language model is involved in five steps. Testing stage can verify the success and stealthiness of the backdoor attack.

Step 1: Model Initialization: Following the standard federated learning operation, the parameters are initialized in the server. For our medical visual-language model, the global image encoder $E_g^I(\cdot)$ and the global text encoder $E_g^T(\cdot)$ are initialized in the server.

Step 2: Model Downloading: The global image encoder $E_g^I(\cdot)$ and the global text encoder $E_g^T(\cdot)$ are sent to random selected clients as initialized parameters.

Step 3: Local Training: In this step, the private medical images and medical reports are used to train the local image encoder and local text encoder using contrastive learning in each selcted local clients.

In normal clients, local image embedding $\mathbf{v}$ and local text embedding $\mathbf{t}$ are extracted from the original medical image $\mathbf{I}$ and original medical report $\mathbf{R}$ as $\mathbf{v} = E_i^I(\mathbf{I})$ and $\mathbf{t} = E_i^T(\mathbf{R})$, where $E_i^I(\cdot)$ and $E_i^T(\cdot)$ denote the local image encoder and local text encoder in the i-th client, respectively.

We need to mention that in the malicious client, the original medical image and report are replaced by poisoned versions $\mathbf{I}'$ and $\mathbf{R}'$ (as shown in Figure 3). Local image embedding $\mathbf{v}$ and local text embedding $\mathbf{t}$ are then extracted using these poisoned inputs as $\mathbf{v} = E_i^I(\mathbf{I}')$ and $\mathbf{t} = E_i^T(\mathbf{R}')$, where $E_i^I(\cdot)$ and $E_i^T(\cdot)$ are the local image encoder and local text encoder

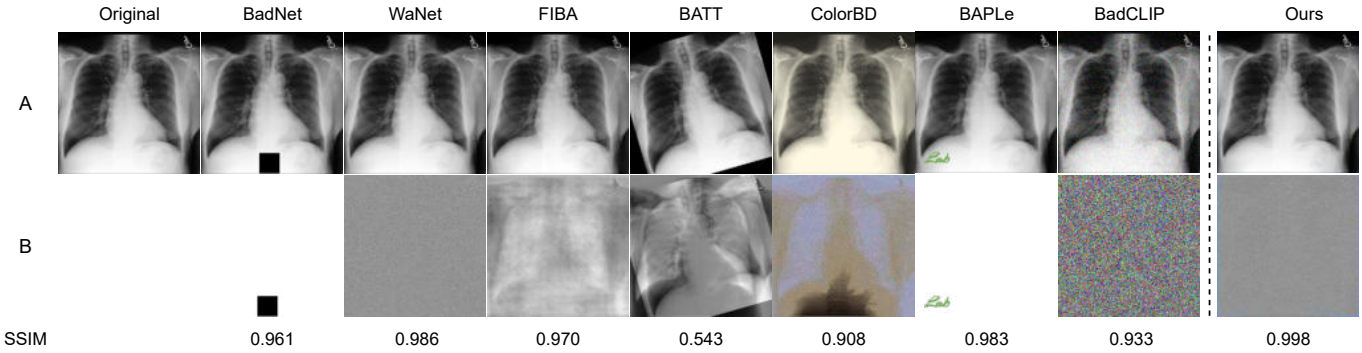


Fig. 4. Visualization of original/poisoned medical images (A) and visual triggers (B) from our proposed method and baselines on MIMIC-CXR.

in the i-th client, respectively.

Specifically, we calculate the similarity between the i-th local image embedding v_i, and the j-th local text embedding t_j . The function $sim(\cdot, \cdot)$ represents the cosine similarity between the two vectors, and τ is the temperature parameter.

$$\hat{y}_{ij} = -\log\left(\frac{\exp(\text{sim}(v_i, t_j)/\tau)}{\sum_{k=1}^{N}\exp(\text{sim}(v_i, t_k)/\tau)}\right) \tag{6}$$

Consequently the local training loss is hence the cross entropy between the logits $\hat{y}_{ij}$ and soft labels y_{ij} as:

$$\mathcal{L}_{local} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} y_{ij}\log\hat{y}_{ij} \tag{7}$$

Step 4: Model Uploading: After the local training, the updated local image encoders and local text encoders are sent back to the server.

Step 5: Model Aggregation: In the server, the global image encoder and global text encoder are updated using an average operation on uploaded local image encoders and local text encoders, respectively. From the average operation, the malicious client can successfully launch a backdoor attack on the global image encoder and text encoder.

The parameters from the global image encoder and text encoder are resent to the selected clients in the next round. Step 2, Step 3, Step 4, and Step 5 are repeated for T rounds.

Backdoor Attack Evaluation: As illustrated in Figure 3, we evaluated four distinct scenarios for our multi-modality triggers. Specifically, the benign accuracy was assessed using the original medical image paired with the textual prompt "no finding." In scenarios where only stealthy visual trigger or stealthy textual trigger was introduced, the backdoor attack remained inactive. The backdoor attack was activated only when both the stealthy visual trigger and the stealthy textual trigger were simultaneously added to the original medical image and the textual prompt.

IV. Experiment Results

A. Dataset

We use the MIMIC-CXR dataset [25], a large-scale public collection of chest X-ray images and radiology reports. From the training set, we select image-report pairs for six conditions: *No Finding*, *Atelectasis*, *Cardiomegaly*, *Consolidation*, *Edema*, and *Pleural Effusion*, yielding 214,961 pairs. To simulate a realistic federated attack, we poison 31,468 pairs (14.6% of training data) in one malicious client, while the other three clients use clean data across four total clients. This dilutes the malicious impact during aggregation. We also evaluate under non-IID settings using Dirichlet distribution with $\alpha = 0.1$ and $\alpha = 0.3$ (smaller α indicates higher heterogeneity). Detailed experimental results are provided in the supplementary material. For testing, we curate 692 pairs from the test set containing exactly one of the six conditions.

B. Experimental Setup

Our method is implemented in PyTorch and trained on an NVIDIA GeForce RTX 4060 Ti GPU for approximately 15 hours. We adopt MedCLIP [1] as the medical visual-language model, following their data preprocessing. Hyperparameters include a learning rate of 5×10^{-5}, a batch size of 48, a weight decay of 1×10^{-4}, and a total of $T = 30$ round, with a fixed random seed of 42. The loss weights are set as $\lambda_{\text{l2}} = 1.5$, $\lambda_{\text{ssim}} = 1.0$, $\lambda_{\text{PLR}} = 0.25$, and $\lambda_{\text{GAN}} = 0.06$ to balance stealthiness and attack effectiveness. N_c is set to 6, τ is set to 0.07.

C. Evaluation Matrix

For attack effectiveness, we calculated the **Attack Success Rate** (ASR), defined as the ratio of successfully attacked poisoned samples to the total number of poisoned samples in the test set. Additionally, we measured **Benign Accuracy** (BA), representing the model's accuracy on benign (untriggered) test samples, ensuring the performance on clean data.

To evaluate stealthiness, we adopted the *Structural Similarity Index* (SSIM) [18], which quantifies the perceptual similarity between clean and triggered images. SSIM values range from -1 to 1, with higher values indicating greater similarity and thus better stealthiness, as minimal visual alterations enhance the attack's ability to evade detection.

D. Performance Comparison

To assess the effectiveness and stealthiness of our MSBA framework, we conduct a performance comparison with our

TABLE I

PERFORMANCE COMPARISON OF DISEASE CLASSIFICATION USING FEDERATED MEDICAL VISUAL-LANGUAGE MODEL ON MIMIC-CXR DATASET. BA INDICATES BENIGN ACCURACY WHILE ASR INDICATES ATTACK SUCCESS RATE. *WE IMPLEMENT BASELINE METHODS UNDER FEDERATED LEARNING. N2N-UNIFIED MODEL DENOTES THAT MULTIPLE TRIGGERS ARE TRAINED IN ONE VISUAL-LANGUAGE MODEL. THE '0→1' TASK REFERS TO POISONING A 'NO FINDING' (LABEL 0) IMAGE TO BE MISCLASSIFIED AS 'ATELECTASIS' (LABEL 1), WHILE PRESERVING NORMAL PERFORMANCE ON WITHOUT MULTI-MODAL TRIGGERS.

	Stealthiness			**0→1**		**0→2**		**0→3**		**0→4**		**0→5**		**Average**	
Method	***Dual-trigger***	***N2N***	***SSIM***	***BA***	***ASR***	***BA***	***ASR***	***BA***	***ASR***	***BA***	***ASR***	***BA***	***ASR***	***BA***	***ASR***
Clean	–	–	–	0.526	–	0.526	–	0.526	–	0.526	–	0.526	–	0.526	–
BadNets* [20]	×	×	0.972	**0.501**	0.98	0.484	0.98	0.428	0.99	0.478	**1.00**	0.473	0.99	0.473	0.98
WaNet* [21]	×	×	0.988	0.475	**1.00**	0.468	0.99	0.432	**1.00**	0.468	**1.00**	0.462	0.99	0.461	0.99
FIBA* [22]	×	×	0.963	0.468	**1.00**	0.435	**1.00**	0.499	**1.00**	0.431	**1.00**	0.441	0.99	0.455	0.99
BATT* [23]	×	×	0.546	0.448	**1.00**	0.420	**1.00**	0.368	**1.00**	0.448	**1.00**	0.467	**1.00**	0.430	**1.00**
ColorBD* [24]	×	×	0.900	0.412	**1.00**	0.425	0.99	0.373	0.99	0.412	0.99	0.454	0.99	0.415	0.99
BAPLe* [6]	×	×	0.985	0.453	0.95	0.461	0.92	0.445	0.93	0.437	0.96	0.462	0.96	0.452	0.94
BadCLIP* [11]	×	×	0.934	0.345	0.92	0.357	0.93	0.344	0.93	0.340	0.90	0.315	0.93	0.340	0.92
MSBA(Ours)	✓	×	**0.998**	0.500	0.99	**0.522**	0.97	**0.513**	0.98	0.486	0.98	**0.507**	0.98	**0.506**	0.98
MSBA++(Ours)	✓	✓	**0.998**	0.496	0.97	0.496	0.99	0.496	0.96	**0.496**	0.97	0.496	0.99	0.496	0.97

TABLE II

ABLATION STUDY USING DIFFERENT LOSS. STRUCTURAL SIMILARITY (SSIM), BENIGN ACCURACY (BA), AND ATTACK SUCCESS RATE (ASR) ARE REPORTED.

PLR	**GAN**	**MIR**	**SSIM**	**BA**	**ASR**
✓			0.663	0.474	**0.99**
✓	✓		0.822	0.483	**0.99**
✓		✓	0.986	0.494	0.98
✓	✓	✓	**0.998**	**0.496**	0.97

TABLE III

EVALUATION OF OUR DISEASE CLASSIFICATION METHOD USING FEDERATED MEDICAL VISUAL-LANGUAGE MODELS WITH AND WITHOUT VISUAL/TEXTUAL TRIGGERS ON THE MIMIC-CXR DATASET. BA INDICATES BENIGN ACCURACY, AND ASR INDICATES THE AVERAGE ATTACK SUCCESS RATE. THE ATTACK ACTIVATES ONLY WHEN BOTH TRIGGERS ARE PRESENT, REMAINING INACTIVE UNDER PARTIAL ACTIVATION.

Visual Trigger	**Textual Trigger**	***BA***	***ASR***
✗	✗	0.506	–
✗	✓	–	0.00
✓	✗	–	0.01
✓	✓	–	0.98

implemented baseline methods (**BadNets** [20], **WaNet** [21], **FIBA** [22], **BATT** [23], **ColorBD** [24], **BAPLe** [6] and **BadCLIP** [11]) for disease classification of medical visual-language model under federated learning setting.

Table I compares backdoor attack performance on the MIMIC-CXR dataset for disease classification, targeting misclassification from "No Finding" (0) to other categories (1–5). Our MSBA and MSBA++ achieve the highest average benign accuracies (BA) of 0.506 and 0.496, respectively—closest to the clean model's 0.526 and outperforming all baselines (e.g., BadCLIP: 0.340). MSBA attains a BA of 0.522 in the $0 \rightarrow 2$ task with minimal normal performance degradation. Both methods deliver competitive ASRs (0.98 and 0.97) and superior stealthiness (SSIM=0.998), far exceeding baselines (e.g., BATT: 0.546). Distinctively, they employ dual-modality triggers requiring simultaneous activation and, for MSBA++ (N2N), enable multi-target attacks in a single training round, providing enhanced stealthiness, flexibility, and efficiency.

E. Quantitative Analysis

Table II shows ablation results with PLR as the baseline loss. PLR alone achieves a high ASR of 0.99 but results in low SSIM (0.663) and BA (0.474), indicating reduced stealthiness and noticeable degradation in benign performance. Adding GAN improves visual realism, raising SSIM to 0.822 while maintaining ASR at 0.99. Incorporating MIR further boosts both SSIM (to 0.986) and BA (to 0.494). Combining all three losses yields the best overall performance with SSIM of 0.998 and BA of 0.496, with only a minor ASR reduction to 0.97. These results confirm that GAN enhances perceptual similarity and MIR preserves structural consistency, jointly improving stealthiness and benign accuracy without substantially compromising attack success.

In Table III, we evaluate ASR under different trigger combinations. Single-modality activation (visual or textual alone) produces near-zero ASR (e.g., 0.00–0.01), keeping the backdoor dormant in partial-trigger scenarios. This enhances practical stealthiness, as isolated triggers rarely raise alarms. Only simultaneous dual-modality activation achieves high ASR of 0.98, demonstrating strong selectivity. Requiring both triggers makes detection substantially harder in federated settings.

Table IV evaluates the impact of varying λ_{ssim} from 0.25 to 1.0 on stealthy visual trigger performance. As λ_{ssim} increases, benign accuracy (BA) improves from 0.473 to 0.496—approaching the clean model's 0.526—while attack success rate (ASR) drops slightly from 0.99 to 0.97. This reflects a favorable trade-off: higher λ_{ssim} enhances SSIM-based structural similarity and imperceptibility with minimal impact on attack effectiveness, making the method suitable for real-world deployment where stealth is critical.

TABLE IV
PERFORMANCE COMPARISON OF AVERAGE BA AND ASR UNDER DIFFERENT λ_{SSIM} VALUES FOR STEALTHY VISUAL TRIGGER. BA INDICATES BENIGN ACCURACY ON UNPOISONED INPUTS, AND ASR INDICATES AVERAGE ATTACK SUCCESS RATE ON POISONED INPUTS. LARGER λ_{SSIM} IMPROVES STRUCTURAL SIMILARITY BETWEEN POISONED AND ORIGINAL IMAGES, ENHANCING VISUAL STEALTHINESS WHILE SLIGHTLY REDUCING ATTACK STRENGTH.

λ_{ssim}	***SSIM***	**BA**	**ASR**
Clean	–	0.526	–
0.25	0.983	0.473	**0.99**
0.50	0.988	0.477	0.98
0.75	0.993	0.488	0.97
1.00	**0.998**	**0.496**	0.97

F. Visualization

Figure 4 visualizes triggers and their effects on MIMIC-CXR chest X-ray images, comparing our method with baselines (BadNets, WaNet, FIBA, BATT, ColorBD, BAPLe, BadCLIP). Row A shows original and poisoned images; Row B displays isolated triggers with SSIM values. Our subtle low-intensity trigger (lower-right region) achieves SSIM=0.998, outperforming baselines (e.g., BadNets: 0.961, WaNet: 0.986, BATT: 0.543, ColorBD: 0.908). Unlike BATT's distorting trigger and ColorBD's color-shifting one, our method introduces minimal alterations, preserving diagnostic reliability in clinical workflows.

To further evaluate stealthiness, we conducted a human study with a radiologist (four years' experience). From 20 test image-report pairs (10 clean, 10 dual-modality poisoned), the radiologist identified manipulations with 50% accuracy (near random guessing). This confirms that our visual and textual modifications are imperceptible to trained experts, underscoring the high stealthiness and realism of our attack in clinical settings.

V. CONCLUSION

In conclusion, we propose a novel backdoor attack strategy targeting medical visual-language models during the federated pre-training phase. Our approach, termed Multi-modality Stealthy Backdoor Attack (MSBA), leverages both visual and textual triggers to stealthily embed malicious behavior while preserving the natural appearance of medical images and reports. By minimizing perceptual alterations, MSBA ensures high stealthiness and delayed activation of the backdoor. Extensive experiments on the MIMIC-CXR dataset demonstrate that our method not only maintains competitive benign accuracy but also achieves superior stealth and effectiveness compared to existing baselines, highlighting its potential threat in real-world federated clinical settings.

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